The price of standing still: stationary impostors for emergent information metrics

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August 15, 2026

How well can a stationary state impersonate the time-averaged mutual-information metric of a moving quantum state — and what stationary resources does the impersonation require? We answer this across fixed-point, quasiperiodic, chaotic, metastable, localized, and driven (kicked-Ising, DTC-like) spin-chain dynamics. First, time averaging and metric construction do not commute: the diagonal ensemble, whose reduced density matrices reproduce the infinite-time averages, misses the time-averaged metric by 43–90%, because mutual information is nonlinear in the state. Second, the repair is family-resolved: thermal-window ensembles suffice for chaotic classes (miss ~ 0.04, consistent with eigenstate thermalization); a generalized Gibbs ensemble suffices for generic integrable quenches (~ 0.15) but fails few-mode quasiperiodic states (~ 0.5); population-only descriptions of the quasiperiodic metric fail smoothly-in-energy (~ 0.32, 0/80 runs) or at best scrape the stationarity threshold (sparse and fully unrestricted, ~ 0.23–0.24 at N=10); stationary states with coherences inside degenerate energy blocks match it almost exactly (miss ~ 0.005). Every class thus has a near-exact stationary impostor; the classes differ in the *stationary resource* the impostor needs — temperature, charges, fine-tuning, or degenerate-block coherence. Third, weak dephasing *pins* the moving quasiperiodic metric to its time average while destabilizing chaotic-class metrics, singling out the same class dynamically. The study is a registered design with a public, commit-hashed record; six claim-level corrections — four internal, two prompted by external review — are reported as results.

1 Introduction

A stable emergent structure can stand in two different relations to the substrate that carries it. It can be *merely instantiated*: some motionless configuration of the substrate would carry the same structure, and the observed dynamics is incidental to it. Or it can be *actively maintained*: the structure exists only because the substrate moves, and no frozen configuration reproduces it. The distinction is easy to state and surprisingly hard to test — most diagnostics of "underlying dynamics" do not actually certify that anything is moving, and most claims of "the dynamics sustains the structure" are never confronted with a serious search for a motionless impostor.

This paper builds that test and runs it, in the most controlled setting we could construct: finite spin chains, where the emergent structure is the mutual-information-graph metric of Refs. [1, 2, 3] — pairwise mutual information converted to link weights and shortest-path distances on a posited site factorization — and the substrate dynamics ranges over fixed-point, quasiperiodic, chaotic, metastable, localized, and driven (kicked-Ising, DTC-like [4, 5, 6]) classes. Prior work on such information-metric structures has been static: ground-state network robustness under projective attacks [2], metricity as a phase diagnostic [3]. Our object is the *time dependence* of the structure over a moving state, and its need — or lack of need — for that motion.

The test has three probes. A *witness battery* certifies that the state moves while the metric does not, under a requirement we treat as non-negotiable: a witness of motion must vanish identically on a frozen state. A *coherence-removal response* measures what happens to the metric when the state is projected onto its motionless diagonal ensemble. And a *motionless-comparator search* optimizes explicitly over stationary states for one that carries the same time-averaged metric. The study is a registered design — every metric, threshold, and analysis rule fixed before code was written, every run seeded from committed manifests, every post-hoc change a dated amendment (Appendix C) — because a study of this shape has one dominant failure mode, self-deception through flexible metric choice, and we preferred to make our mistakes in public.

The mistakes were made, caught, and are reported as results. Six times, a correction mechanism fired: a registered class certificate was misspecified (Poisson statistics for a free chain); the out-of-time-order correlator, which we had registered as a witness of state motion, fired on a frozen ground state — the null test correctly classifying it as an operator-spreading diagnostic [7] rather than a state-motion witness; an early robustness comparison was exposed by our adversarial re-analysis as a floored-denominator artifact; a narrow first version of the comparator search over-claimed unmatchability for two classes that a hardened search then matched; and two successive rounds of external review each forced the comparator search open further — first to unrestricted diagonal populations, which overturned our remaining existential claim, then to stationary states with coherences inside degenerate energy blocks, which matched the last resisting class almost exactly. What survives this gauntlet is correspondingly compact:

1. **An operational test** of instantiated-vs-maintained emergent structure (null-silent witnesses, coherence removal, motionless-comparator search), portable to any system with a computable structural functional.

2. **A non-commutation result.** Time averaging and metric construction do not commute: the diagonal ensemble — the motionless state whose two-site reduced density matrices are the infinite-time averages [8] — misses the time-averaged metric by 43–90%, and the mismatch is not a finite-window artifact: the exact window-averaged state misses by nearly the same amount (Sec. 6).

3. **A resource-resolved matching spectrum.** Every class's time-averaged metric has a near-exact stationary impostor, but the stationary *resource* the impostor requires is class-resolved: thermal windows suffice for the chaotic classes ($\sim$3 parameters, miss $\sim$ 0.04, as eigenstate thermalization suggests [9]); a generalized Gibbs ensemble [8, 10] matches the generic integrable quench but not the few-mode quasiperiodic states; every population-only family — smooth in energy (0/80 runs at $\sim$ 0.32), sparse, or fully unrestricted — leaves the quasiperiodic class straddling the threshold at N=10; stationary states with coherences inside degenerate energy blocks match it almost exactly (miss $\sim$ 0.005). The class discriminator is the *smooth-in-energy representation gap* of the metric, not the existence of an impostor.

4. **A sign structure.** Weak dephasing *stabilizes* the moving metrics of coherence-carried classes (negative log drift ratio, flat over two decades of noise rate) while destabilizing chaotic-class metrics — the class that is hardest to match is the one noise defends. The mechanism is standard decoherence damping [11]; the class-resolved geometric reading appears, within the scope of our survey [2, 3, 12, 13], to be unreported.

5. **A methods record.** The null-silence filter, the registered/amended/pre-committed staging of sharp statistical criteria with a measured seed-lottery episode, and the complete dated amendment log as Appendix C.

The driven track adds an instructive case study: the DTC-like regime's metric survives removal of its drive unchanged — localization, not the subharmonic motion, holds it (Sec. 7).

What this paper does not claim. The metric is an information-theoretic construct on a posited factorization; its factorization dependence is measured (Sec. 8) and it is not spacetime geometry — no claim here touches gravity or holography. No mechanism is claimed as new physics. No assumption is made that information is ontologically primitive. All dynamics is stated against an external laboratory clock. We make no priority claims about preregistration in computational physics; the artifact is offered, not its firstness. The model family is small ($N \leq 14$) and the conclusions are in-model.

Section 2 defines the family, functional, witnesses, and protocol; Secs. 3–7 report results; Sec. 8 states the scope walls; Sec. 9 discusses what the test opens up.

2 Models, functional, witnesses, and protocol

2.1 Model family

All models are open chains of N qubits with $\hbar = 1$ and the nearest-neighbour coupling setting the unit of energy and time. Every dynamical statement below is made with respect to an external laboratory clock; the drive of the Floquet family defines a second, stroboscopic clock derived from it. Nothing in this study bears on the question of clocks internal to a system, which we regard as out of scope.

We use three tracks. Track A collects four closed-system classes chosen to span the qualitative types of quantum motion (Table 1).

Track C reuses the chaotic and quasiperiodic Hamiltonians as "scrambling" and "integrable" regimes with Néel-anchored product-state ensembles, and adds a localized regime (XXZ with random fields in the strong-disorder, MBL-like phenomenology [14]: $\Delta = 1$, $W = 8$; 20–40 disorder realizations; $\langle r \rangle = 0.38$–$0.40 \in [0.36, 0.42]$). Track B is a kicked-Ising discrete time crystal modeled on the established protocol [4, 5]: an imperfect global π-flip (ε the imperfection) alternating with disordered Ising interactions, 20 disorder realizations $\times$ 5 initial product states, with two comparator regimes — no interactions (r1) and no disorder (r2).

The class certificates are *preregistered sanity checks*. One failed as originally registered: we specified a Poisson level-statistics window for the integrable class, which is a heuristic for generic integrable models and is inapplicable to a free chain whose many-body spectrum is exactly the subset-sums of its single-particle spectrum. The replacement certificate — reconstruction of the many-body spectrum from single-particle energies to 10^{-8} — is strictly stronger, and the misspecification plus its dated repair are part of the study record (Appendix C). A second certificate finding: the quasiperiodic incommensurability requirement is *exhaustively unsatisfiable* at $N = 8$ (0 of 56 magnon triples pass), which fixed the usable size range $(10, 12)$ for class comparisons.

2.2 The information metric and its stationarity

From the state (pure or mixed) we compute all one- and two-site reduced density matrices, the pairwise mutual information $I_{ij} = S_i + S_j - S_{ij}$, and normalize $x_{ij} = I_{ij}/(2 \ln 2)$. Following the emergent-metric construction of Ref. [1], link weights are $w_{ij} = -\ln x_{ij}$, regularized by a floor $x_{\min} = 10^{-6}$, and the *metric* $D(t)$ is the matrix of weighted shortest-path distances on the complete graph over sites (Fig. 1). We emphasize the scope of the word "metric" here: D is the metric structure of the mutual-information graph on a *posited* site factorization. It is not a claim about spacetime, and its dependence on the factorization is measured, not assumed away (Sec. 8).

Stationarity is judged on the window $\mathcal{W} = [20, 200]$ (units of inverse coupling; stroboscopic periods for Track B), sampled every $\Delta t = 0.5$: with $\bar{D}$ the window mean and $\delta\Phi(t) = \|D(t) - \bar{D}\|_F / \|\bar{D}\|_F$, the metric is stationary iff $\max_{t \in \mathcal{W}} \delta\Phi(t) < \varepsilon_\Phi = 0.25$. The threshold was set by a preregistered calibration pilot that measured the construction's finite-size fluctuation floor (the originally registered 0.05 sat below the noise floor of every dynamical class — an instrument artifact the pilot existed to catch; Appendix C, Amendment 3). A cap diagnostic — the same drift restricted to above-floor links — is reported alongside every verdict to expose any cap dominance; none of the verdicts below is cap-dominated.

2.3 Witnesses and the null-silence requirement

The claim "the metric is stationary while the state moves" is only as good as the certificate of motion. We require every *witness of motion* to satisfy a null-silence condition: **it must vanish identically on a frozen state.** The null comparator is class (i) — an eigenstate, whose evolution is a global phase — and the preregistered rule is that any witness which fires on the null is discarded as a witness, whatever its other virtues.

The battery: (W1) the participation ratio PR_A of the binned Bohr spectral measure

$$A(\omega) = \sum_{m,n} |c_m|^2 \, |c_n|^2 \, \delta(\omega - |E_m - E_n|), \quad (1)$$

class	Hamiltonian	initial ensemble	class certificate
(i) fixed point	transverse-field Ising, $H = -\sum_i \sigma_i^z \sigma_{i+1}^z - g \sum_i \sigma_i^x$, $g = 1.5$	ground state (single deterministic run)	gapped, nondegenerate
(ii) quasiperiodic	XX chain, $H = \frac{1}{2} \sum_i (\sigma_i^x \sigma_{i+1}^x + \sigma_i^y \sigma_{i+1}^y)$	20 superpositions of 3 single-magnon modes, pairwise-incommensurate Bohr gaps	free-spectrum reconstruction $< 10^{-8}$
(iii) chaotic	mixed-field Ising, $(h_x, h_z) = (0.9045, 0.8090)$	20 Haar-random product states	$\langle r \rangle = 0.529$–$0.542 \in [0.51, 0.55]$
(iv) metastable	ferromagnetic Ising, $g = 0.05$, dressings $\delta g_i \in [-0.01, 0.01]$	$\lvert\uparrow \cdots \uparrow\rangle$, 20 dressings	quasi-degenerate doublet

Table 1: Track-A classes. Certificates are preregistered sanity checks, run at every system size used before any confirmatory analysis.

equal to 1 exactly on the null; (W2) the recurrence distance $d(t) = 1 - |\langle \psi(t_0) | \psi(t) \rangle|^2$, identically zero on the null, summarized by its window mean; (W4) the off-diagonal energy coherence $\Xi = \sum_{E_m \neq E_n} |\rho_{mn}|^2$, computed over energy-*distinct* pairs so that coherence within a degenerate level — which does not move — does not fire it; and, for Track B, (W5) the subharmonic Fourier weight of the stroboscopic magnetization at half the drive frequency, with its rigidity curve $h_{\rm sub}(\varepsilon)$. All are invariant under global phase and basis relabelings in the senses catalogued in Appendix D.

We also registered (W3) an out-of-time-order correlator, $C(r,t) = \frac{1}{2} \langle |[\sigma^z_{i_0+r}(t), \sigma^z_{i_0}]|^2 \rangle$, with saturation value and arrival time as statistics. The null test disqualified it for *our specific purpose* — measured, not argued: on the frozen fixed point it rises to $C \approx 1.11$ by $t = 1.5$. This is not a defect of the OTOC. OTOCs are understood in the primary literature as diagnostics of Heisenberg-operator spreading [7], which a gapped ground state supports, and firing on an eigenstate is expected behaviour for such an object. The point the null test makes is narrower and, we think, worth making: an operator-spreading diagnostic is not a witness of *state* motion, and the two roles are easy to blur when "dynamics" is used informally. Under the registered rule its statistics were removed from the class-separation criterion, with consequences reported in Sec. 4; it remains in our outputs in its proper role.

2.4 Comparators and perturbation protocols

Three controls give the sustained-by test its content. The *diagonal ensemble* $\bar{\rho} = \sum_n |c_n|^2 |n\rangle\langle n|$ is the canonical motionless partner of a run [8, 9]: exactly stationary, and — for a nondegenerate spectrum — carrying the *infinite-time* averages of all reduced density matrices. Two qualifications matter and both are measured in Sec. 6: our windows are finite (the exact window-averaged state is compared against $\bar{\rho}$ directly), and the XX-based spectra are degenerate, so the infinite-time average retains coherences inside degenerate energy blocks that $\bar{\rho}$ discards — a distinction that becomes decisive there. We registered $\bar{\rho}$ as "matched to the time-averaged metric by construction" — which is false, and measurably so: mutual information is nonlinear in the (correctly time-averaged) reduced density matrices, and $\Phi[\bar{\rho}]$ sits 43–90% away from $\bar{D}$ depending on class. The corrected comparator methodology — an explicit optimization over stationary states — is described with its results in Sec. 6. The *coherence-removal test* projects the state onto its diagonal ensemble mid-window (energy-basis dephasing; populations are conserved, so the result is exactly $\bar{\rho}$) and asks whether the metric's behaviour changes. We emphasize what this operation is not: it is not switching the Hamiltonian off — setting $H = 0$ would freeze the current state and its metric in place — it removes energy-basis coherence, i.e. the motion, while keeping the time-averaged local structure. The driven track uses a genuinely different operation, *drive removal* (Sec. 7). And three *perturbation protocols* — a local Hamiltonian quench $\lambda\sigma^z$, a local

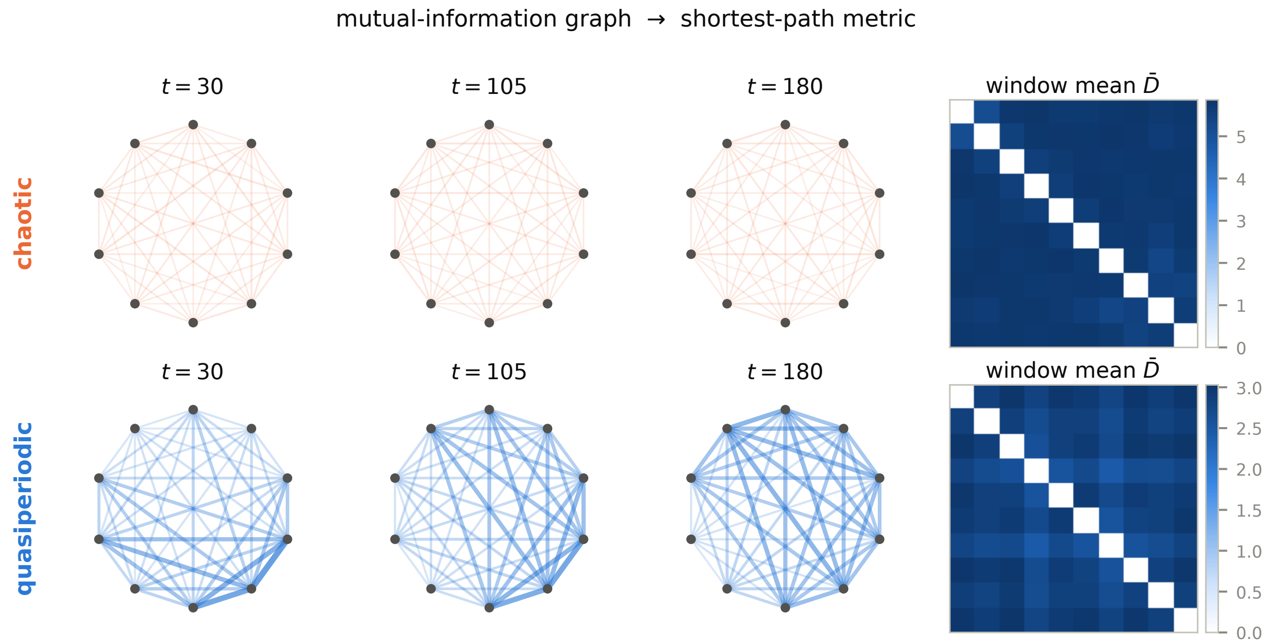

Figure 1: The pipeline, on real data (representative $N = 10$ runs). Nodes are sites; edge weight is pairwise mutual information; the metric D is the shortest-path distance matrix. Top: a chaotic-class state — the graph barely changes across the analysis window (its weak, near-uniform mutual information yields the large, static distances of the window mean $\bar{D}$). Bottom: a quasiperiodic state — the graph visibly rearranges as the incommensurate magnon beats evolve.

dephasing channel of rate γ, and the loss of two non-adjacent sites — probe robustness through the scale-free log drift ratio

$$\log\rho \;=\; \log\left[\frac{\max_{t>t_p}\delta\Phi_{\mathrm{pert}}(t)}{\max\big(\max_{t>t_p}\delta\Phi_{\mathrm{unpert}}(t),\,10^{-3}\big)}\right], \tag{2}$$

with strengths $(\lambda, \gamma) = (0.1, 0.01)$ fixed by the calibration pilot before any confirmatory run.

2.5 Preregistration protocol and statistics

The study's workflow follows the Shared Substrate protocol for sustained human–AI research collaboration [15], whose central rule — no result exists until it is recorded in a versioned, canonical substrate — generates the audit trail this paper reports. Concretely: every metric, threshold, witness, comparator, and analysis rule above was fixed in a specification document before implementation code was written; every run consumed seeds from manifests committed to the repository before execution; and every post-hoc change is a dated entry in an amendment log, reproduced in full as Appendix C with commit hashes. The study ran as: specification → threshold review → calibration pilot (exploratory, excluded from confirmatory statistics) → confirmatory campaign → an adversarial companion analysis tasked with attacking our own conclusions → a witness-battery reformalization forced by the null-test failure, itself confirmed on fresh seeds. Class separation is judged by exact Mann–Whitney AUC ≥ 0.95 for at least two witness statistics per class pair at two system sizes (singleton classes by exact-value exclusion); robustness differences by $|\Delta\,\overline{\log\rho}| > \ln 1.5$ with disjoint BCa bootstrap confidence intervals, replicated with consistent direction at two sizes; disordered ensembles resample at the realization level. Where a sharp threshold met a finite ensemble, seed sensitivity was measured and stabilized ($n = 40$, Sec. 4); where it met a third size, the trend was checked descriptively ($N = 14$, Appendix B).

3 Results I: stationary metrics over witnessed motion

The baseline campaign establishes the phenomenon the rest of the paper interrogates: metrics that do not move over states that demonstra-

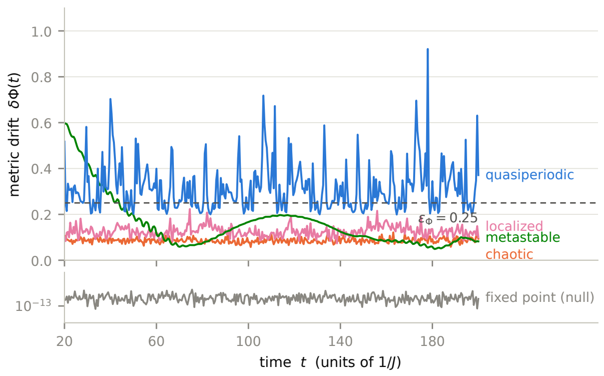

Figure 2: Metric drift $\delta\Phi(t)$ for representative runs ($N = 10$). The chaotic metric is stationary under the preregistered threshold while the quasiperiodic metric oscillates far above it; the metastable class shows its slow coherent wave; the localized run sits near the threshold (the class straddles it, 80/100 stationary at $N = 12$). Bottom strip: the frozen null at machine precision. Ensemble verdicts are given in the text.

bly do (Fig. 2).

At the calibrated threshold $\varepsilon_\Phi = 0.25$, the chaotic and scrambling ensembles are metric-stationary in every run (20/20 at each size; max window drift 0.09–0.21) while their motion witnesses read near maximum: $\Xi \geq 0.96$, recurrence mean $\approx 1 - 1/D$, and Bohr participation ratios in the hundreds to thousands. The driven track is the sharpest instance: at drive imperfection $\varepsilon = 0.03$, all 100 runs are metric-stationary while the subharmonic witness reads 0.939 — a locked period-doubled magnetization response over a frozen mutual-information metric. At the other extreme, the quasiperiodic and metastable classes have genuinely moving metrics (max drift 0.53–1.61; no run stationary), the integrable regime likewise (0.35–0.64), and the fixed point is stationary to machine precision (2×10^{-13}) with every witness silent — the null behaving as a null. The localized regime straddles the threshold (80/100 stationary at $N = 12$, drift 0.14–0.34) and is treated as a boundary case throughout.

Two honesty notes frame everything that follows. First, for the chaotic classes, metric stationarity over motion is *expected*: the metric is built from two-site observables, and the equilibration of local observables in quenched chaotic systems is standard physics. The existence table is the setup, not the result. Second, none of these verdicts is an artifact of the weight floor in the metric construction: the above-floor drift diagnostic tracks the full-graph drift in every class, and the mean graphs have no floor-dominated links.

4 Results II: the witness discipline at work

This section reports the study's methodological core: what the preregistered witness requirements did when the data pushed back. We present it as a dual record — the original registration failed; the ratified repair passed on fresh seeds — because both halves are results (Fig. 3).

The OTOC is not a witness of motion. The null-silence requirement (Sec. 2.3) demands that a witness vanish on the frozen class-(i) state. The out-of-time-order correlator fails: on the gapped ground state it reaches the arrival threshold at $t^* = 1.5$ and saturates at $C \approx 1.11$. Nothing is wrong with the OTOC as physics — it certifies operator spreading, which a frozen state supports — but that is precisely why it cannot certify that a *state* is moving. The preregistered discard clause executed, removing both OTOC statistics from the class-separation criterion.

The discard had teeth. With the OTOC gone, the original battery left exactly one class pair under-witnessed: scrambling vs. localized separated on Ξ alone (AUC 0.985/0.990 at the two sizes), with the Bohr participation ratio degrading with size ($0.89 \to 0.56$) and the registered recurrence statistic — the window *minimum* — landing at 0.71/0.945. Criterion (a), as originally registered, therefore **fails**. The counterfactual is recorded: had the OTOC been retained, its arrival time separates that pair at AUC 1.0 (localized fronts never arrive; scrambling fronts arrive at $t \approx 2$). The failure is single-cause, and it is a finding about witness design: the natural scrambling/localization discriminator is not null-silent, and the null-silent battery we registered was one statistic short.

The repair, and what fresh seeds taught us. The reformalized battery replaces the recurrence minimum — structurally uninformative for slow classes, since a slowly rotating state revisits its reference — with the recurrence *mean*, which is null-silent by construction and class-resolving in practice. On the data that motivated the redesign, all 18 pair-size checks passed; we labeled

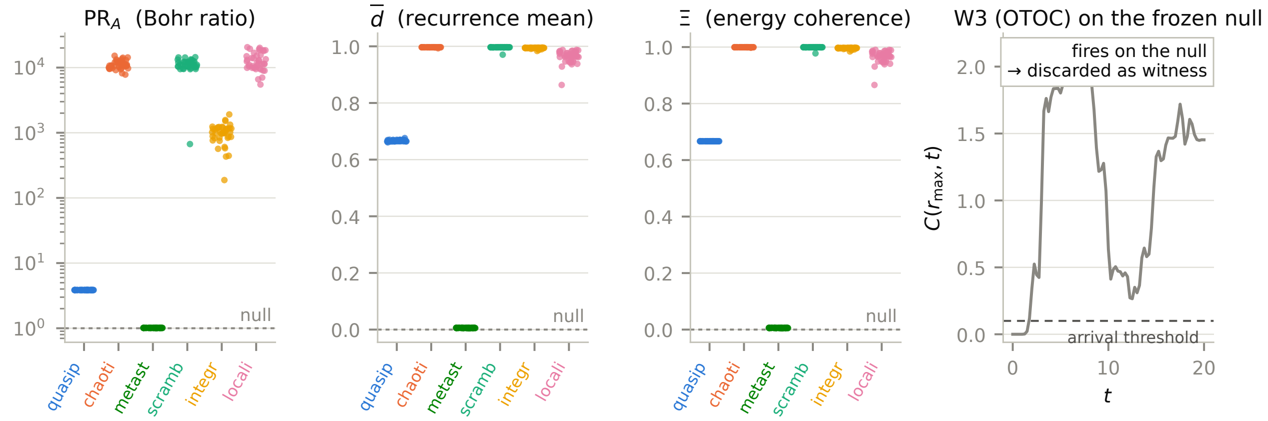

Figure 3: The final witness battery (fresh seeds, $n = 40$, $N = 12$; per-run values, one dot per run). Dashed lines: the frozen null's exact values. All three statistics are null-silent; class separation is visible directly. Right panel: the registered witness W3 (OTOC) evaluated on the frozen null — it crosses its own arrival threshold at $t^* = 1.5$ and saturates near 1.1–1.5, firing on a state that does not move; under the preregistered null-silence clause it was discarded from the criterion.

that table validation, not confirmation, and re-adjudicated on fresh committed seeds. At the registered ensemble size ($n = 20$) the fresh seeds **failed** a different pair — scrambling vs. integrable at $N = 10$, all three statistics between 0.85 and 0.93 — while passing at $N = 12$. The diagnosis is statistical, not physical: a sharp AUC threshold of 0.95 estimated from 20-vs-20 samples carries a standard error of 0.05–0.08, so near-threshold verdicts are seed-lotteries. Doubling the ensembles ($n = 40$, fresh seeds again) stabilized every margin: **all 18 pair-size checks pass**, the previously marginal pair at 0.952–0.979, and the repaired pair at 0.987–0.992 (the Bohr ratio for scrambling vs. localized remains honestly weak, 0.59 at $N = 12$ — two statistics carry the pair, as the criterion requires). A descriptive third size ($N = 14$, computed from eigenbasis coefficients alone) confirms the monotone trend: all three statistics ≥ 0.976.

To keep the epistemic status of each stage unambiguous, Table 2 separates them explicitly; only the first row carries the original registration's status, and the final verdict is *pre-committed* (fresh seeds, outcome accepted in advance either way, under ratified amendments) rather than preregistered.

The two methodological claims we take from this: null-silence is a nontrivial filter that separates state-motion witnesses from operator-spreading diagnostics; and sharp-threshold criteria on small ensembles must be fresh-seeded and, near the threshold, power-stabilized — our original-seed validation table would have over-claimed.

stage	status	outcome
original battery, registered seeds	registered	(a) FAILS
reformalized battery, original seeds	exploratory	18/18 (validation only)
reformalized, fresh seeds, n=20	pre-committed	FAILS (marginal pair)
reformalized, fresh seeds, n=40	pre-committed	**PASSES 18/18 (final)**

Table 2: Stages of the class-separation criterion. Amendments between stages are dated in Appendix C.

5 Results III: class-resolved robustness and the dephasing sign structure

The robustness criterion asks whether metric stationarity responds to perturbation differently in different dynamical classes. Under local dephasing at the calibrated rate $\gamma = 0.01$, it does, with a sign structure that is the study's most striking single measurement (Fig. 4).

The mean log drift ratios, replicated at two sizes with disjoint confidence intervals and consistent direction: chaotic and scrambling +2.1 to +2.2 (noise strongly destabilizes their stationary metrics), localized +1.5, integrable +0.75 — and the coherence-carried classes *negative*:

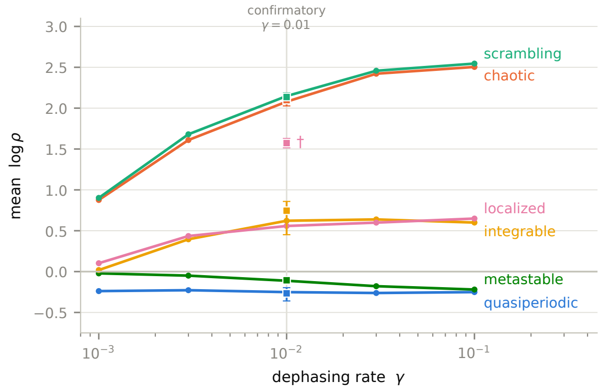

Figure 4: Dephasing response by class: mean $\log \rho$ versus rate γ (descriptive sweep, curves; first 10 manifest runs per class, Néel-primary for the localized regime). Squares with intervals: the confirmatory single-γ ensemble means with BCa 95% CIs. (†: the confirmatory localized ensemble uses 5 initial states per realization and reads higher than the Néel-primary sweep — an ensemble-composition difference, not a discrepancy.) The sign structure — negative for coherence-carried classes, strongly positive for chaotic ones — is stable over two decades and never reorders.

metastable $-0.09/-0.11$, quasiperiodic -0.27. Weak dephasing makes the quasiperiodic metric *more* stationary than its own unperturbed dynamics: the noise damps the coherences whose beating carries the metric's oscillation, pinning $D(t)$ to its time average. Every pairwise contrast required by the registered criterion clears the $\ln 1.5$ threshold in both tracks at both sizes; the class ordering (chaotic $\approx$ scrambling $>$ localized $>$ integrable $>$ metastable $\gtrsim$ quasiperiodic, the last two negative) is the pilot's ordering, confirmed at scale.

A descriptive strength sweep shows the sign structure is a regime, not a point: across two decades $\gamma \in [10^{-3}, 10^{-1}]$ the quasiperiodic response is flat at ≈ -0.25, the metastable response strengthens monotonically ($-0.02 \rightarrow -0.22$), the chaotic-class response rises and saturates ($+0.9 \rightarrow +2.5$), and the ordering never reorders. The mechanism is standard open-system physics — decoherence damping of coherence-carried oscillations, the weak-rate side of the Zeno family [11] — and we claim none of it as new. The class-resolved *geometric* reading — noise as a stabilizer of moving information metrics and a destabilizer of chaotic ones, with the sign tracking the dynamical class — is, to the extent of our survey [2, 3, 12, 13], unreported; we state this conditionally and note the survey was not systematic.

The other two protocols are clean nulls at the registered strengths: the local quench leaves every dynamical class's drift ratio at $|\overline{\log \rho}| \leq 0.07$, and two-site loss reads 0.07–0.25 everywhere. (The fixed point's large quench ratio, $+4.4$, is floor-referenced — its unperturbed drift is exactly zero — and carries no class information; instrument note in Appendix B.) All robustness statements are statements at the calibrated strengths and, per the sweep, within the probed regime — not strength-independent class properties.

6 Results IV: the motionless-comparator search — existence versus resources

The sharpest form of the maintained-vs-instantiated question is: *does there exist a motionless state carrying the same time-averaged metric — and if so, what does it take to build one?* An important caveat frames this whole section: for the classes whose metric is genuinely moving (quasiperiodic, metastable, integrable), the object under discussion is the *time-averaged* metric $\bar{D}$, which is a stationary summary by construction; only for the metric-stationary classes is $\bar{D}$ also the metric one would observe. This section reports the search, including the four corrections that shaped it — two forced by our own audits, two by successive rounds of external review (Fig. 5, Table 3).

The canonical comparator is not matched. The diagonal ensemble $\bar{\rho}$ — the motionless state carrying the run's infinite-time averaged two-site reduced density matrices (up to the degenerate-block coherences discussed below) — was registered as metric-matched by construction. It is not: mutual information is nonlinear in the reduced density matrices, and $\Phi[\bar{\rho}]$ sits 43–90% of $\|\bar{D}\|$ away from the run mean, depending on class. Relatedly, our first robustness comparison against $\bar{\rho}$ produced a large "the comparator is more fragile" differential that an adversarial re-analysis exposed as a floored-denominator artifact (the comparator has no baseline drift to compare against; its absolute perturbed response is in fact *smaller*). Both corrections are dated in the study record; what survives them is stronger

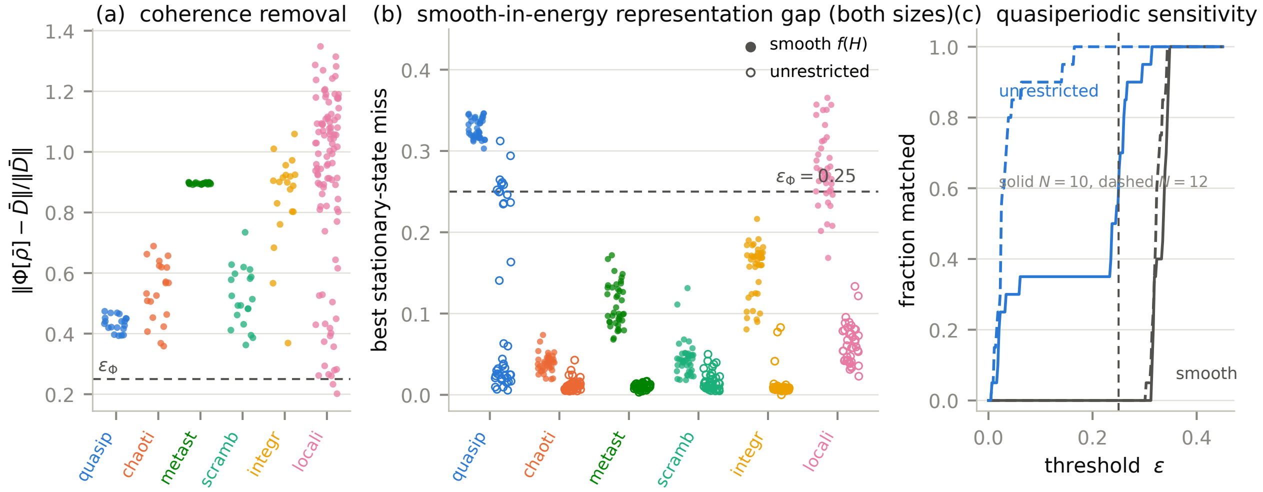

Figure 5: (a) The coherence-removal measurement: projecting the state onto its diagonal ensemble mid-window moves the metric by 43–90% of $\|\bar{D}\|$ in every class — removing the motion changes the metric. (b) The smooth-in-energy representation gap: best miss found, $\|\Phi[\sigma]-\bar{D}\|/\|\bar{D}\|$ per run, both sizes pooled, for smooth-$f(H)$ stationary ensembles (filled; ≤ 24 parameters) versus unrestricted diagonal-population optimization (rings; 2^N parameters). Only the quasiperiodic class shows a robust, all-run gap between the two; Table 3 completes the family spectrum — with coherences inside degenerate energy blocks, the quasiperiodic metric too is matched almost exactly. (c) Threshold sensitivity for the quasiperiodic class: fraction of runs matched versus threshold, per search family and size; the registered $\varepsilon_\Phi = 0.25$ (vertical line) sits in the steep region of the unrestricted $N = 10$ curve, so counts at that threshold are sensitive — the smooth/unrestricted gap is not.

than what they replaced.

The coherence-removal measurement. Projecting onto $\bar\rho$ mid-window moves the metric by 43–90% of $\|\bar{D}\|$ (chaotic 0.54, scrambling 0.52, integrable 0.85, metastable 0.90, localized 0.90, quasiperiodic 0.43). Removing the motion does not freeze the metric in place; it changes the metric. For the metric-stationary classes this registers as a drift increase with disjoint confidence intervals.

Finite windows versus infinite time. $\bar\rho$'s reduced density matrices are *infinite*-time averages; our windows are finite. The distinction is measured rather than assumed small: on one representative run per class and size we computed the exact window-averaged state $\rho_{\mathcal{W}} = |\mathcal{W}|^{-1}\int_{\mathcal{W}} \rho(t)\,dt$ and compared its metric with $\Phi[\bar\rho]$. In the relaxing classes the two nearly coincide (metric gap 0.003–0.23 across classes and sizes) and miss $\bar{D}$ by nearly the same amount (quasiperiodic: 0.466 versus 0.465 at N=10) — the non-commutation result is about averaging versus nonlinearity, not about window length. The one exception is instructive: the N=10 metastable run's slow doublet does not dephase within the window, so $\rho_{\mathcal{W}}$ retains near-degenerate coherence that $\bar\rho$ discards and lands far closer to $\bar{D}$ (0.08 versus 0.90); at N=12, where the doublet physics differs, both misses are small (≈ 0.09).

The staged search. Stationarity means $[\sigma, H] = 0$. The search opened in five stages of increasing freedom, over full ensembles at both criterion sizes, each widening forced by an audit or a review round: (i) three natural families (thermal at both temperature signs, depolarized diagonal ensemble, Gaussian microcanonical; ≤ 2 parameters each); (ii) a smooth-$f(H)$ family (log-populations parameterized by Chebyshev polynomials in the scaled energy, orders to $K = 96$, multi-start); (iii) — prompted by the first review round, which correctly observed that stage (ii) is not all diagonal states — an *unrestricted* optimization over all 2^N populations (softmax parameterization, analytic gradient through the entropies and shortest paths, verified against finite differences, multi-start), together with sparse population vectors on $k \leq 32$ eigenstates; (iv) a generalized Gibbs ensemble over the XX chain's free-fermion mode occupations [8, 10] (N Lagrange multipliers), the canonical stationary description of an integrable quench; and (v) — prompted by the second review round, which correctly observed that even stage (iii) is not all stationary states — optimization over *block-coherent* stationary states. The point behind stage (v) is a spectral fact worth stating plainly: the XX-based spectra are massively degenerate (N=10: 243 distinct energies in a 1024-dimensional space, largest block 32; N=12: 729 in 4096, largest 64), and $[\sigma, H] = 0$ forces σ to be block-diagonal only across *distinct* energies — within a degenerate block it may carry arbitrary coherence, exactly the coherence the infinite-time average itself retains. Stage (v) therefore spans the full commutant ($\sum_B |B|^2$ real parameters: 7776 at N=10, 46656 at N=12), with a per-run validity check against stage (iii), which the block family strictly contains (Appendix A). Two earlier over-claims are part of the record: the family-only stage wrongly suggested metastable and integrable were unmatchable (corrected at stage ii), and stage (ii) wrongly suggested the quasiperiodic class was categorically unmatched (corrected at stages iii–v). All misses reported below are the best found by nonconvex optimization — upper bounds, not certified minima.

The final result is a **resource split, not an existence split**. Table 3 gives the full price list for the quasiperiodic class, the one whose metric resists longest; the other classes are cheaper at every stage. Within smooth-$f(H)$ ensembles — whose parameter count does not grow with dimension — chaotic and scrambling metrics are matched almost exactly in every run (median 0.035–0.046; a Gaussian microcanonical window suffices, as eigenstate thermalization would suggest [9]), metastable (0.09–0.13) and integrable (0.16–0.17) follow, and localized is marginal (median $\approx$ 0.27). The generalized Gibbs ensemble — the stationary description integrable systems are supposed to relax to [8, 10] — matches the generic integrable quench in every run (median 0.14 at N=10, 0.17 at N=12) but fails the quasiperiodic states outright (0/20 at both sizes, medians 0.63 and 0.47): a few-mode superposition is precisely an initial condition whose mode-occupation description does not fix its pair correlations. For the quasiperiodic class, every *population-only* description struggles: **the smooth-$f(H)$ search leaves it matched in 0 of 80 runs** (medians 0.34/0.32 at N=10/12, robust across mode numbers $m = 3, 4, 5$), and raising the Chebyshev order from K=2 to 96 moves the miss only from 0.36 to 0.32 — a plateau, so the failure is not parameter starvation within smoothness. Sparse supports and the fully unrestricted diagonal search do better but at best scrape the calibrated threshold at N=10 (medians 0.23–0.24), an order of magnitude above the match quality every other class enjoys; at N=12 the unrestricted search matches (20/20, median 0.025), partly a parameter-counting effect — 2^N free populations chasing $N(N-1)/2$ target entries. **Block coherence changes the picture qualitatively**: the commutant search matches every quasiperiodic run at both sizes essentially exactly (20/20 each; medians 0.0052 at N=10 and 0.0035 at N=12, best 10^{-9}), at the match quality of the cheap impostors of other classes, and the integrable class likewise (0.0077–0.0080). Degenerate-block coherence — not population fine-tuning — is the stationary resource the quasiperiodic metric actually requires.

Two independent instruments single out the same class, in this corrected sense: the quasiperiodic metric carries the largest *smooth-in-energy representation gap* — the distance between its best smooth-$f(H)$ reproduction and its best-found stationary reproduction, 0.32 versus 0.005, against 0.04 versus 0.01 for chaotic — and it is the only metric that weak noise *stabilizes* (Sec. 5). Fig. 5(c) shows the smooth-versus-unrestricted gap is insensitive to the threshold choice even where counts at ε_Φ are not. Whether some principled complexity measure on stationary states makes the representation gap precise — parameter count is a crude proxy, and the block-coherent result shows the relevant resource can be a *kind* of coherence rather than a number of parameters — we pose as the open question this study leaves sharpest. Conversely, the chaotic classes' stationary metrics are straightforwardly compatible with their motion: the simplest thermal impostor carries the same structure, while the robustness signature of Sec. 5 still distinguishes the classes dynamically.

7 Results V: the driven regime

The driven track — a kicked-Ising protocol in its rigid subharmonic (DTC-like) regime [4, 5, 6] — supplies the study's cleanest stationary-with-witness regime and its most instructive compatible-with verdict (Fig. 6). We say "DTC-

stationary family	free parameters (N=10)	median best miss (runs $< \varepsilon_\Phi$) $N = 10$	$N = 12$
natural families (thermal, depolarized, microcanonical)	≤ 2	0.38 (0/20)	0.37 (0/20)
generalized Gibbs (free-fermion charges)	N	0.63 (0/20)	0.47 (0/20)
smooth $f(H)$ (Chebyshev, best over $K \leq 96$)	≤ 96	0.32 (0/20)	0.32 (0/20)
sparse support (best over $k \leq 32$ eigenstates)	≤ 32	0.23 (19/20)	—
unrestricted diagonal	2^N	0.24 (12/20)	0.025 (20/20)
block-coherent commutant	$\sum_B \lvert B\rvert^2$ (7776)	0.0052 (20/20)	0.0035 (20/20)

Table 3: The price of standing still, for the class where it is highest: best stationary reproduction found for the quasiperiodic time-averaged metric, by family. Every entry is an upper bound attained by non-convex search; entries differing at the 0.01 level (the sparse row's best-over-support lands slightly below the unrestricted row's five-start result) reflect optimizer variance, the two-order-of-magnitude drop of the last row does not. The N=12 smooth entry uses $K \leq 24$.

like" deliberately: at $N \leq 14$ with 20 disorder realizations we observe locked subharmonic response and a rigidity curve consistent with the discrete-time-crystal phase, but we perform no size or lifetime scaling and make no phase claim [6].

At $\varepsilon = 0.03$, deep in the rigid regime, all 100 runs are metric-stationary while the subharmonic witness reads 0.939: a period-doubled magnetization response, locked across 20 disorder realizations, above a mutual-information metric that does not move. Stationarity degrades with drive imperfection (88/100 at $\varepsilon = 0.06$, 52/100 at 0.10), tracking the witness. The rigidity curve is measured over its full range: the subharmonic weight holds above 0.65 through $\varepsilon = 0.20$, crosses 1/2 at $\varepsilon_c \approx 0.23$ at our drive parameters, and collapses by $\varepsilon \approx 0.45$ — a complete critical-strength measurement for this realization of the protocol of Refs. [4, 5].

The drive-removal test resolves the maintained-vs-instantiated question for this regime, and resolves it *against* the drive: removing the kicks at mid-window collapses the witness ($0.95 \to 0.00$) while the metric's stationarity persists and slightly improves (mean post-window drift $0.110 \to 0.086$). The residual static disordered evolution, not the drive, holds this metric — the witnessed subharmonic motion is compatible with the stationary structure, not required by it. We registered this as an open question with exactly this outcome listed as reportable, and report it accordingly. Consistently, the perturbation protocols at the calibrated strengths barely move the DTC-like regime's metric ($\overline{\log \rho} = 0.01$–$0.19$).

The comparator regimes behave asymmetrically, and honestly. Without interactions (r1), the subharmonic peak is destroyed at $\varepsilon = 0.03$ (0.13) — the fine-tuned control fails as expected, certifying that interactions produce the rigidity. Without disorder (r2), however, the clean interacting drive does *not* thermalize on any horizon we probed: its subharmonic response persists undiminished (0.89–0.90) to 2000 periods. At small ε the r2 regime is not a usable thermalizing control, and the discriminating comparator burden falls entirely on r1 — a comparator-scope finding we report rather than bury.

8 Scope walls and limitations

The metric belongs to the factorization. Everything here is computed on the natural site partition, and the dependence on that choice is not small and not uniform: under fixed nonlocal two-site frame changes, the metric moves by a class-dependent amount — mean relative change 0.11 (chaotic), 0.15 (scrambling), 0.20 (localized), 0.22 (integrable), 0.42 (quasiperiodic), 0.53 (metastable, maximum observed 0.82). The coherence-carried classes — including the class with the largest representation gap in Sec. 6 — have the most frame-dependent metrics. This does not undermine the within-frame comparisons (every matchability and robustness statement compares objects in the same partition, which transforms both sides together), but it bounds the interpretation: these are properties of the metric-on-a-factorization, not of the state alone. Notably, strictly local (single-site) frame changes leave the metric invariant to machine precision — the dependence lives entirely at the partition level, which is where the construction's

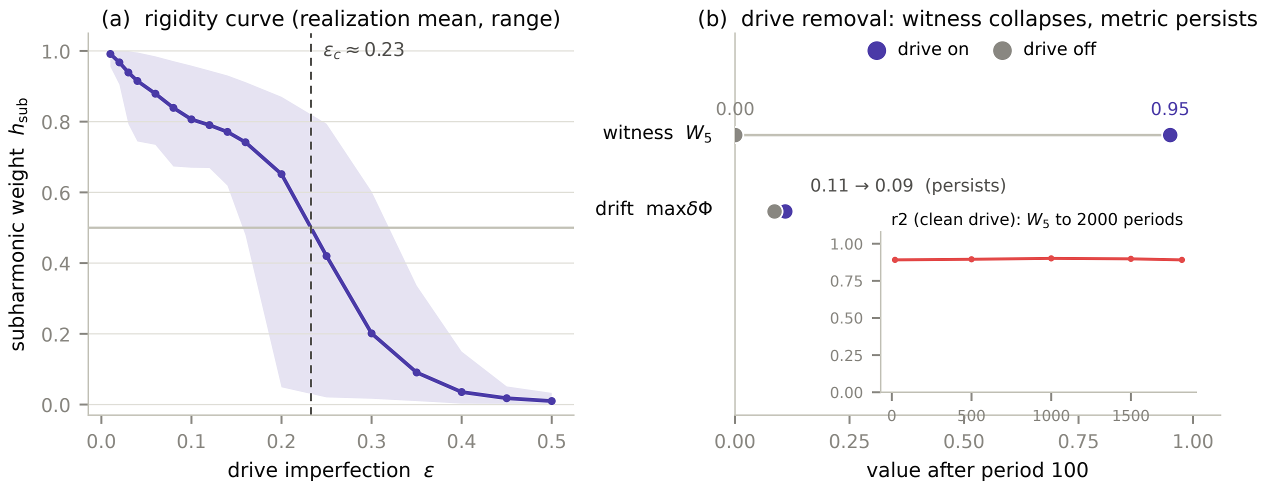

Figure 6: (a) The full rigidity curve of the driven track: subharmonic weight $h_{\rm sub}$ versus drive imperfection (realization mean and range, seed-paired across ε), crossing $1/2$ at $\varepsilon_c \approx 0.23$ (interpolated). (b) The drive-removal test at $\varepsilon = 0.03$: removing the kicks collapses the subharmonic witness while the metric drift persists essentially unchanged — localization, not the drive, holds this metric. Inset: the clean interacting comparator (r2) does not thermalize on any probed horizon; its subharmonic weight is undiminished at 2000 periods.

own authors located their caveat [1].

Finite size, finite family, finite search. Criterion verdicts are established at $N \in \{10, 12\}$ with a descriptive $N = 14$ point; the quasiperiodic construction is exhaustively unsatisfiable at $N = 8$ (0/56 candidate mode triples), which is a warning about small-size instantiations of incommensurability generally. The stationarity threshold $\varepsilon_\Phi = 0.25$ is calibrated to this construction's finite-size noise floor and has no significance beyond it. The comparator search spans parametric families, smooth functions of H, a generalized Gibbs ensemble, sparse and unrestricted diagonal populations, and the full block-coherent commutant; every reported miss is the best found by non-convex optimization, so "not matched" statements are bounded by the search, not proofs of nonexistence — though the block-coherent stage leaves no class unmatched for such a proof to concern. The localized regime straddles the stationarity threshold (80/100) and we have kept it out of every headline count. Robustness statements hold at the calibrated strengths and within the swept regime ($\gamma \in [10^{-3}, 10^{-1}]$). All clocks are external laboratory clocks; nothing here addresses systems that must supply their own.

9 Discussion and outlook

What the study delivers is not a mechanism but an instrument, and a measured answer narrower than the question in our title. Is the emergent metric motion-borne? Not in the existential sense we initially hoped to establish: every class's time-averaged metric, including the quasiperiodic one, has a near-exact stationary impostor once the full commutant — coherences inside degenerate energy blocks included — is searched. What the measurements do establish is threefold: time averaging and metric construction do not commute (the canonical motionless partner misses by 43–90%, and the exact window-averaged state misses by the same amount); the *stationary resource* a faithful impostor requires is sharply class-resolved — a temperature window for the thermalizing classes, conserved charges for the generic integrable quench, degenerate-block coherence and nothing less for the quasiperiodic class; and weak noise stabilizes exactly the class whose metric no smooth-in-energy ensemble represents. "Motion-borne" survives as a motivating question and as an open conjecture about *smooth, low-resource* stationary descriptions — not as a demonstrated existential conclusion.

The sharpest open question the data leaves is the representation gap itself: is there a principled complexity measure on stationary states — one that prices block coherence as well as parameter count, since the block-coherent impostor shows the decisive resource can be a kind of coherence rather than a number of parameters — under which the quasiperiodic time-averaged metric is provably expensive, and does the gap survive larger systems, where parameter counting increasingly favours fine-tuned matches? A second question follows the sign structure: is there a form of robustness that *recurrent* or quasiperiodic dynamics provides and fixed points cannot? The coincidence reported here — the hardest-to-match class is the one noise defends — is the evidence such a mechanism would leave, and the driven track supplies a controlled arena in which to look (with the caution of Sec. 7: the most spectacular witnessed motion in this study turned out *not* to hold its metric).

The test template itself is not specific to spin chains. Any system with a computable structural functional over a dynamical substrate admits the same three probes: certify the motion with null-silent witnesses, remove the coherence that carries it, and search honestly for a motionless impostor — reporting the impostor's complexity, not just its existence. Candidate applications suggest themselves wherever stable emergent structure rides on flux — resting-state functional connectivity over neural dynamics, metabolic network structure over flux-carrying steady states, stability of ecological or market structure over turnover — and we offer the template for these settings as a proposal only: nothing beyond spin chains has been demonstrated here. The distinction the test operationalizes — structure that merely exists versus structure that is actively kept in existence — is, of course, much older than physics; we make no attempt to engage that literature here.

On the methods side, we draw two conclusions we believe generalize. Null tests for witnesses are cheap and clarifying — the OTOC episode suggests that diagnostics informally credited with certifying "dynamics" deserve routine auditing against frozen states, if only to fix which kind of dynamics they certify. And a registered design with committed seeds changed the outcome of this study six times — four corrections from our own audit mechanisms and two from successive rounds of external review; the amendment log (Appendix C) is our answer to the question of what the substrate discipline [15] buys in a purely computational setting.

The obvious open front is scale and covariance: whether any version of this question can be asked of structures that deserve the word "geometry" with fewer scare quotes — partition-covariant functionals, larger systems, internally clocked dynamics — remains exactly as open as it was, and none of the present results shortens that road. What they do establish is that the road's first step is measurable: already in twelve qubits, one can ask what a structure's cheapest motionless impostor costs, and get a class-resolved answer.

Acknowledgements. The numerical study — implementation, execution, analysis, and the adversarial companion — was carried out with Claude (Anthropic) as AI research assistant, operating under the Shared Substrate protocol [15] whose record forms Appendix C. The programme's initiating knowledge base was drafted in collaboration with OpenAI's ChatGPT. An external pre-submission review prompted the unrestricted comparator analysis and several framing corrections reported here. All rulings, decisions, and the final text are the author's responsibility.

Funding. This work received no external funding; it was carried out in the author's free time, for the joy of it.

Data and code availability. The full study record — specification, seed manifests, amendment log, evidence packets, session logs, analysis code, and all run outputs — is available in the public repository `https://github.com/olegroshka/ideg`, pinned at the submission tag (`paper-v1`).

A Numerics

Models and distributions. Track-A/C Hamiltonians and parameters are fully specified in Table 1 and Sec. 2.1; disorder draws are $h_i \sim U[-W, W]$ with $W = 8$ for the localized regime and $\delta g_i \sim U[-0.01, 0.01]$ for the metastable dressings. The driven track evolves by the Floquet unitary $U_F = e^{-iH_2t_2}\, e^{-iH_1t_1}$ with $t_1 = t_2 = 1$, $H_1 = \frac{\pi}{2}(1-\varepsilon)\sum_i \sigma_i^y$ and $H_2 = \sum_i J_i\, \sigma_i^z\sigma_{i+1}^z + \sum_i h_i\sigma_i^z$, $J_i \sim U[0.05, 0.15]\pi$, $h_i \sim U[0, \pi]$ (r1 sets $J_i = 0$; r2 sets $h_i = 0$);

disorder and initial states are seed-paired across ε and regimes.

Dephasing channel. The perturbation protocol applies local σ^z dephasing at rate γ: Lindblad generator $\mathcal{D}[\rho] = \gamma \sum_i (\sigma_i^z \rho\, \sigma_i^z - \rho)$, implemented by Trotter splitting (step 0.5) of the exact unitary and the exact damping mask $e^{-2\gamma\, \Delta t\, d_H(a,b)}$ on the computational-basis coherences, d_H the Hamming distance — exact for the commuting pure-dephasing part, splitting error $O(\Delta t^2)$.

Comparator optimization. The stationary search minimizes $\|\Phi[\sigma(p)] - \bar{D}\|_F / \|\bar{D}\|_F$ over eigenbasis populations p. Stages: fixed families (grids over β, depolarization λ, and Gaussian window (E_0, σ_E)); smooth $f(H)$ ($\log p = \sum_{k<K} c_k T_k(\tilde{E})$, $K \leq 24$, Powell, multi-start, ≤ 800 evaluations per start); unrestricted ($p = \mathrm{softmax}(\theta)$, $\theta \in \mathbb{R}^{2^N}$, L-BFGS-B with the analytic gradient assembled from $\partial S/\partial p_n = -\mathrm{tr}[R_n(\ln \rho + 1)]$ per reduced density matrix, $\partial w/\partial x = -1/x$ off the weight floor, and shortest-path edge subgradients from predecessor-tracked Floyd–Warshall; five starts including the best smooth solution; gradient verified against finite differences, relative deviation 1.7×10^{-3}, dominated by path-tie subgradients). Convergence: L-BFGS-B to $\|g\| < 10^{-8}$ or 400 iterations; multi-start scatter across the five starts is small compared to the class gaps reported.

GGE and block-coherent stages. The generalized Gibbs ensemble is built on the XX chain's exact free-fermion eigenbasis (single-particle modes ϵ_k from the hopping matrix, many-body eigenstates as mode-occupation configurations; the basis is self-checked against direct diagonalization, $\max |HV - VE| < 10^{-8}$), with populations $p_S \propto \exp(-\sum_{k \in S} \lambda_k)$ over occupied-mode sets S and the N multipliers λ_k optimized by L-BFGS-B with numerical gradients. The block-coherent stage parameterizes the commutant as $\sigma = \bigoplus_B A_B A_B^\dagger / Z$ with one complex factor A_B per degenerate energy block B (positivity and stationarity by construction), warm-started from the square root of the best diagonal solution; the objective's analytic gradient is assembled from per-pair, per-block cross tensors of eigenstate reduced density matrices, and each run is validity-checked against the unrestricted diagonal stage: the block family strictly contains the diagonal family, so a block result above a diagonal one signals optimizer shortfall, not family structure. The check caught a sign error in the objective during development, and in the two-start production data it flagged exactly one integrable run, where the diagonal search's five starts found a 10^{-8} optimum the block search did not (0.02). A start-parity rerun of the full block stage (five starts — warm, two jitters, two random — matching the diagonal stage's count) resolved the flag as pure optimizer shortfall: the flagged run dropped to 3×10^{-10}, below the diagonal value, while every quasiperiodic optimum was reproduced unchanged at both sizes (medians 0.0052/0.0035; largest single-run change 0.0012) — the warm start dominates, and Table 3 is start-count-insensitive. All reported misses remain found optima, not certified minima. Sparse supports re-run the unrestricted optimizer restricted to the k largest diagonal-ensemble populations, $k \in \{1, \ldots, 128\}$; entries above $k = 32$ show optimizer regression (a harder landscape, not a better bound) and are excluded from Table 3. The finite-window comparison integrates $\rho(t)$ over $\mathcal{W}$ by trapezoidal quadrature at the sampling grid and evaluates the same metric functional on the result.

Diagonalization and evolution. Exact diagonalization throughout (dense to $N = 14$, dimension 16384); pure-state evolution via eigendecomposition, density-matrix evolution for the dephasing protocol via Trotter splitting of the unitary and the exact per-step damping mask (step 0.5), $N \leq 10$. State-norm drift across all unitary runs $< 4 \times 10^{-15}$; trace drift of dephased runs recorded per run at the same scale. Python 3.11 / NumPy 2.3 / SciPy 1.15; every random draw seeded from committed manifests; total compute across pilot, confirmatory, adversarial, and hardening campaigns of order a machine-day on a 24-core workstation. The $N = 14$ witness point uses the identity $|\langle \psi(t_0) | \psi(t) \rangle|^2 = |\sum_n p_n e^{-iE_n(t-t_0)}|^2$, so recurrence statistics require only eigenbasis coefficients. The comparator search precomputes per-eigenstate two-site reduced density matrices, making each candidate stationary state's metric a population-weighted sum evaluable in milliseconds at any dimension.

B Statistics

Class separation: exact Mann–Whitney AUC, symmetrized $\max(A, 1-A)$ since direction is not registered; threshold 0.95; ≥ 2 statistics per pair; two sizes; singleton classes (the fixed point) by exact-value exclusion (the ensemble range must exclude the deterministic value). Robustness: mean log drift ratio per class, BCa bootstrap (1000 resamples, 95%), disjoint intervals plus $|\Delta| > \ln 1.5$, replicated with consistent direction; disordered ensembles resampled at the realization level. Seed sensitivity: at $n = 20$-vs-20 the AUC standard error near $A \approx 0.9$ is 0.05–0.08, so verdicts within one standard error of threshold are lotteries — measured directly here when fresh seeds flipped a marginal pair (Sec. 4); $n = 40$ halves the standard error and stabilized every margin. Size trend for the marginal pair (AUC at $N = 10 \to 12 \to 14$): Bohr ratio $0.979 \to 0.979 \to 1.000$; recurrence mean $0.952 \to 0.975 \to 0.976$; energy coherence $0.956 \to 0.975 \to 0.976$. Instrument notes: the log-ratio floor (10^{-3}) makes exactly-stationary objects read $\log \rho \approx \log(\delta\Phi_{\text{pert}}/10^{-3})$ — class-(i) quench values are floor-referenced and excluded from class inference; the metric's $-\ln x$ weight cap amplifies machine-epsilon mutual-information jitter by up to $x_{\min}^{-1} = 10^6$, setting an irreducible $\sim 10^{-9}$ numerical floor on metric-space identity checks for near-product states.

C The amendment log

The complete dated record of every post-registration change, each with its reason and repository commit; no entry alters data already taken.

D Invariance battery

Every witness statistic and the metric were checked under: global phase; consistent local basis change (state, Hamiltonian, and operators together); local basis change of the state alone; and site reflection. Witness statistics and mutual-information matrices are identical within 10^{-10} in all mandatory items across every class and size (the participation ratio judged relative to its own magnitude, which reaches 10^5–10^6 at $N = 12$). Metric-space deviations carry the cap-amplified numerical floor described in Appendix B and are reported, not thresholded. The state-only local change leaves the metric invariant to machine precision — single-site entropies are local-frame invariant — which localizes all factorization dependence at the partition level; the partition probe of Sec. 8 (fixed random entangling two-site frames on three disjoint pairs, 24–72 samples per class) is the quantitative version.

E Driven-track implementation notes

Disorder realizations and initial states are seed-paired across drive strengths and comparator regimes (identical couplings and fields at equal seed), so rigidity curves and regime contrasts are paired comparisons. The subharmonic weight is computed on an even number of stroboscopic periods so the period-doubled line is the exact Nyquist bin. Drive removal evolves under the interaction Hamiltonian alone in lab time, sampled at the same period boundaries (drive removed, clock kept). The dephasing protocol applies the exact per-period damping mask with the rate stated in lab-time units ($T_{\text{period}} = 2$).

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date	entry	commit
08-11	Spec registered; threshold review (Am. 1): criterion-(b) instrument $\rightarrow$ log drift ratio + calibration pilot	24eaa858ec60 7e40011d56c8
08-11	Am. 2 (pre-run): integrable certificate Poisson $\rightarrow$ free-spectrum reconstruction (mis-specification measured: free chain)	602658b1b96c
08-11	Am. 3 (pilot, owner-ruled): ε_Φ 0.05 $\rightarrow$ 0.25 (below measured noise floor); confirmatory $\lambda = 0.1$, $\gamma = 0.01$	5db69bade460
08-12	Instrument survey: KEEP log-ratio (no change; dated entry)	27ada59e697c
08-12	Confirmatory manifest committed pre-execution; survey window closed at first run	b91ad54da91f
08-12	Addendum 1: quasiperiodic construction unsatisfiable at $N = 8$ (0/56, exhaustive) $\rightarrow$ criterion sizes (10, 12)	a76db0e67e09
08-12/13	Confirmatory executed: W3 fires on null $\rightarrow$ discarded; (a) fails as registered; (b) holds; verdicts of record	fbe324f38f34
08-13	Adversarial companion: fragility-direction retraction (floored denominator); comparator matching assumption refuted	33fdd6287776
08-13	Am. 4 (ratified): battery $\rightarrow \{\mathrm{PR}_A, \overline{d}, \Xi\}$; W3 descriptive; fresh-seed re-adjudication required	9030c0ebdab6
08-13	Fresh seeds $n = 20$: FAIL on a different, threshold-marginal pair (seed lottery measured)	5efaa7309c56
08-13	Am. 5 (ratified): $n = 40$, final verdict accepted either way $\rightarrow$ (a) HOLDS 18/18	3e7c0a787a09
08-13	Hardened comparator search corrects family-only over-claim: single-survivor result	67dda68970b1
08-13	Descriptive hardening sweep: ε_c, 2000-period comparator, γ-grid, partition distribution, $N = 14$	9ce17dec803c
08-14	External pre-submission review received; core criticisms accepted (time-average framing; search-space bound)	—
08-14/15	Unrestricted comparator optimization (review-prompted): every class matchable; smooth-vs-unrestricted naturalness gap replaces the single-survivor claim	075eb96, 2978713
08-15	Major revision: non-commutation thesis; coherence-removal / drive-removal renaming; tempered OTOC framing; DTC-like wording; stages table; self-contained numerics	9c05a1f
08-15	Second review round received; degeneracy criticism accepted: diagonal populations are not all stationary states when the spectrum is degenerate	—
08-15	Block-coherent commutant, GGE, price-curve, and window-gap battery (AR-020d): quasiperiodic matched almost exactly by degenerate-block coherence; resource split replaces complexity split; an objective sign error caught by the containment check (Appendix A)	(this build)

Table 4: The amendment log (12-character commit identifiers in the public repository).